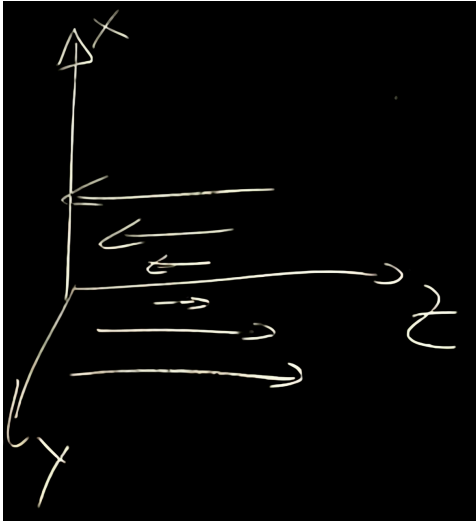


2025-12-18 - Servidio (20)

Harrys Sheeth

PIC grad along x



We assume quasi neutrality, stationarity and translational symmetry

$$n_i \simeq n_e \equiv n; \quad \frac{\partial}{\partial t} = 0; \quad \frac{\partial}{\partial y} = \frac{\partial}{\partial z} = 0$$

so

$$f_{i,e} = f_{i,e}(x, v_x, v_y, v_z)$$

The Vlasov-Maxwell system is

$$\frac{\partial f_\alpha}{\partial t} + \bar{v} \cdot \nabla f_\alpha + \frac{q_\alpha}{m_\alpha} \left(\bar{E} + \frac{\bar{V} \times \bar{B}}{c} \right) \cdot \nabla_{\bar{v}} f_\alpha = 0$$

$$\frac{\partial^2 A_y}{\partial x^2} = -\frac{4\pi}{c} e \left[\int \bar{v} f_i d^3v - \int \bar{v} f_e d^3v \right] \cdot \hat{y}$$

$$\frac{\partial^2 \phi}{\partial x^2} = -4\pi e \left[\int f_i d^3v - \int f_e d^3v \right]$$

We take an integral of the motion: the Energy

$$\frac{d}{dt} \mathcal{E} = 0$$

if f is a function of an invariant of motion, then it is a solution of the Vlasov problem

$$f_\alpha = f_\alpha(\mathcal{E}) \quad \implies \quad \frac{D}{Dt} f_\alpha(\mathcal{E}) = 0$$

f can be any function ($f_\alpha(\mathcal{E}) = \mathcal{E}$, $f_\alpha(\mathcal{E}) = \mathcal{E}^\gamma$ or $f_\alpha(\mathcal{E}) = \exp(\mathcal{E})$).

Proof (bases on the derivation chain rule):

$$\begin{aligned}
\frac{D}{Dt} f_\alpha(\mathcal{E}) &= \frac{\partial f_\alpha(\mathcal{E})}{\partial t} + \bar{v} \cdot \nabla f_\alpha(\mathcal{E}) + \bar{a} \cdot \nabla_{\bar{v}} f_\alpha(\mathcal{E}) = \\
&= \frac{\partial f_\alpha}{\partial \mathcal{E}} \frac{\partial \mathcal{E}}{\partial t} + \frac{\partial f_\alpha}{\partial \mathcal{E}} (\bar{v} \cdot \nabla \mathcal{E}) + \frac{\partial f_\alpha}{\partial \mathcal{E}} (\bar{a} \cdot \nabla_{\bar{v}} \mathcal{E}) = \\
&= \frac{\partial f_\alpha}{\partial \mathcal{E}} \left[\frac{\partial \mathcal{E}}{\partial t} + \bar{v} \cdot \nabla \mathcal{E} + \bar{a} \cdot \nabla_{\bar{v}} \mathcal{E} \right] = \\
&= \frac{\partial f_\alpha}{\partial \mathcal{E}} \left(\frac{D}{Dt} \mathcal{E} \right) = 0
\end{aligned}$$

Now we use

$$\frac{\partial}{\partial y} = \frac{\partial}{\partial z} = 0$$

from which we obtain two new invariants (using the same reasoning)

$$P_y = mV_y + \frac{q}{c} A_y(x) = \text{constant}$$

$$P_z = mV_z = \text{constant}$$

so we can write

$$f_\alpha = f_\alpha(\mathcal{E}, P_y, P_z)$$

Harrys used to use some combinations of the invariants

$$\left\{ \begin{array}{l} \mathcal{E} = \frac{1}{2} m_\alpha (V_x^2 + V_y^2 + V_z^2) + q_\alpha \phi(x) \\ P_y = mV_y + \frac{q}{c} A_y(x) \\ P_z = mV_z \end{array} \right. \iff \left\{ \begin{array}{l} \alpha_1^2 = \frac{2}{m} \left(\mathcal{E} - \frac{P_y^2}{2m} - \frac{P_z^2}{2m} \right) \\ \alpha_2 = \frac{P_y}{m} = V_y + \frac{q}{mc} A_y(x) \\ \alpha_3 = \frac{P_z}{m} = V_z \end{array} \right.$$

which all have the dimension of a velocity.

So the f are drifting maxwellians (drifting because there is a mean velocity)

$$\left\{ \begin{array}{l} f_i(x, V_x, V_y, V_z) = \left(\frac{m_i}{2\pi k_B T_i} \right)^{3/2} \exp \left\{ -\frac{m_i}{2k_B T_i} \left[\alpha_1^2 + (\alpha_2 - U_i)^2 + \alpha_3^2 \right] \right\} \\ f_e(x, V_x, V_y, V_z) = \left(\frac{m_e}{2\pi k_B T_e} \right)^{3/2} \exp \left\{ -\frac{m_e}{2k_B T_e} \left[\alpha_1^2 + (\alpha_2 - U_e)^2 + \alpha_3^2 \right] \right\} \end{array} \right.$$

We want to substitute the α_i inside the $f_{i,e}$ in order to show the dependencies on the velocities and the potential ϕ .

First we need $\alpha_j(\bar{V}, \phi, A_y)$

$$\begin{cases} \alpha_1^2 &= \frac{2}{m} \left[\frac{1}{2} m_\alpha (V_x^2 + V_y^2 + V_z^2) + q_\alpha \phi(x) - \frac{(mV_y + \frac{q}{c} A_y)^2}{2m} - \frac{m^2 V_z^2}{2m} \right] = \\ &= \dots = V_x^2 + \frac{2q}{mc} V_y A_y - \frac{q^2}{m^2 c^2} A_y^2 + \frac{2q}{m} \phi \\ \alpha_2 &= V_y + \frac{q_\alpha}{m_\alpha c} A_y(x) \\ \alpha_3 &= V_z \end{cases}$$

so

$$\begin{aligned} f_\alpha(x, V_x, V_y, V_z) &= \left(\frac{m_\alpha}{2\pi k_B T_\alpha} \right)^{3/2} \exp \left\{ -\frac{m_\alpha}{2k_B T_\alpha} \cdot \right. \\ &\quad \cdot \left[V_x^2 + \frac{2q}{mc} V_y A_y - \frac{q^2}{m^2 c^2} A_y^2 + \frac{2q}{m} \phi + \left[V_y + \frac{q_\alpha}{m_\alpha c} A_y(x) - U_\alpha \right]^2 + V_z^2 \right] \Big\} = \\ &= \left(\frac{m_\alpha}{2\pi k_B T_\alpha} \right)^{3/2} n_\alpha \exp \left\{ -\frac{m_\alpha}{2k_B T_\alpha} \left[V_x^2 + (V_y - U_\alpha)^2 + V_z^2 + \frac{2q_\alpha}{m_\alpha} \phi(x) - \frac{2q_\alpha}{m_\alpha c} A_y(x) U_\alpha \right] \right\} \end{aligned}$$

We need to plug this result into Vlasov maxwell

$$\begin{aligned} \frac{\partial^2 \phi}{\partial x^2} &= -4\pi e \left[\int f_i d^3 v - \int f_e d^3 v \right] \\ \frac{\partial^2 A_y}{\partial x^2} &= -\frac{4\pi}{c} e \left[\int \bar{v} f_i d^3 v - \int \bar{v} f_e d^3 v \right] \cdot \hat{y} \end{aligned}$$

So we need the following terms

$$\int f_i d^3 V, \quad \int f_e d^3 V, \quad \int \bar{v} f_i d^3 V, \quad \int \bar{v} f_e d^3 V$$

We start from the first

(isolating the exponential with velocity terms from that with the potential terms)

$$\begin{aligned} n_i(x) &= \int f_i d^3 V = \left(\frac{m_i}{2\pi k_B T_i} \right)^{3/2} n_{i0} \left\{ \int \exp \left[-\frac{m_i}{2k_B T_i} (V_x^2 + (V_y - U_i)^2 + V_z^2) \right] \cdot \right. \\ &\quad \cdot \exp \left[\underbrace{-\frac{m_i}{2k_B T_i} \left(\frac{2q_i}{m_i} \phi(x) - \frac{2q_i}{m_i c} A_y(x) U_i \right)}_{\gamma} \right] d^3 V \Big\} \end{aligned}$$

We call the argument of the second exponent as γ , this term is independent of V_i .

The integral over the velocity is the integral of a shifted gaussian and we already have the solution.

$$\begin{aligned} n_i(x) &= \left(\frac{m_i}{2\pi k_B T_i} \right)^{3/2} n_{i0} \exp(\gamma) \int \exp \left[-\frac{m_i}{2k_B T_i} (V_x^2 + (V_y - U_\alpha)^2 + V_z^2) \right] d^3 V = \\ &= \left(\frac{m_i}{2\pi k_B T_i} \right)^{3/2} n_{i0} \exp(\gamma) \left(\frac{2\pi k_B T_i}{m_i} \right)^{3/2} \end{aligned}$$

The same reasoning works for the second integral, so

$$\left\{ \int f_i d^3V = n_i(x) = n_{i0} \exp \left\{ \frac{q_i}{k_B T_i} \left[\frac{1}{c} A_y(x) U_i - \phi(x) \right] \right\} \right. \\ \left. \int f_e d^3V = n_e(x) = n_{e0} \exp \left\{ \frac{q_e}{k_B T_e} \left[\frac{1}{c} A_y(x) U_e - \phi(x) \right] \right\} \right\}$$

We move on to the second type of integral, this time we split the integral in d^3V into three integrals in dV_x dV_y and dV_z

$$\int \bar{v} f_i d^3V = \int V_y \left(\frac{m_i}{2\pi k_B T_i} \right)^{3/2} n_{i0} \exp(\gamma) \exp \left[-\frac{m_i}{2k_B T_i} (V_x^2 + (V_y - U_\alpha)^2 + V_z^2) \right] d^3V = \\ = \left(\frac{m_i}{2\pi k_B T_i} \right)^{3/2} n_{i0} \exp(\gamma) \int \exp \left[-\frac{m_i}{2k_B T_i} V_x^2 \right] dV_x \cdot \\ \cdot \int V_y \exp \left[-\frac{m_i}{2k_B T_i} (V_y - U_\alpha)^2 \right] dV_y \int \exp \left[-\frac{m_i}{2k_B T_i} V_z^2 \right] dV_z$$

The integrals along x and z are of the form

$$\int \exp \left[-\frac{m_i}{2k_B T_i} v^2 \right] dv = \sqrt{\frac{2\pi k_B T_i}{m_i}}$$

so

$$\left\{ \int \exp \left[-\frac{m_i}{2k_B T_i} V_x^2 \right] dV_x = \sqrt{\frac{2\pi k_B T_i}{m_i}} \right. \\ \left. \int \exp \left[-\frac{m_i}{2k_B T_i} V_z^2 \right] dV_z = \sqrt{\frac{2\pi k_B T_i}{m_i}} \right.$$

For the y component we apply a change of variable

$$V'_y = V_y - U_i; \quad V_y = V'_y + U_i \quad \implies \quad dV_y = dV'_y$$

so (the eliminated integral is null by symmetry)

$$\int V_y \exp \left[-\frac{m_i}{2k_B T_i} (V_y - U_\alpha)^2 \right] dV_y = \\ = \int_{-\infty}^{\infty} V'_y \exp \left[-\frac{m_i}{2k_B T_i} V'^2_y \right] dV'_y + U_i \int_{-\infty}^{\infty} \exp \left[-\frac{m_i}{2k_B T_i} V'^2_y \right] dV'_y = \\ = U_i \sqrt{\frac{2\pi k_B T_i}{m_i}}$$

So, finally, we have

$$\int \bar{v} f_i d^3V = \left(\frac{m_i}{2\pi k_B T_i} \right)^{3/2} n_{i0} \cdot U_i \left(\frac{2\pi k_B T_i}{m_i} \right)^{3/2} \cdot \exp \left\{ -\frac{q_i}{k_B T_i} \left[\frac{A_y(x) U_i}{c} - \phi(x) \right] \right\}$$

and the explicit results for ions and electrons are

$$\begin{cases} \int \bar{v} f_i d^3V = U_i n_{i0} \exp \left[\frac{eU_i}{ck_B T_i} A_y(x) - \frac{e\phi(x)}{k_B T_i} \right] \\ \int \bar{v} f_e d^3V = U_e n_{e0} \exp \left[-\frac{eU_e}{ck_B T_e} A_y(x) + \frac{e\phi(x)}{k_B T_e} \right] \end{cases}$$

Now we can substitute all these results into the Vlasov-Maxwell system

$$(n_e \simeq n_i \equiv n_0)$$

$$\begin{aligned} \frac{\partial^2 \phi}{\partial x^2} &= -4\pi e n_0 \left\{ \exp \left[\frac{eU_i}{ck_B T_i} A_y(x) - \frac{e\phi(x)}{k_B T_i} \right] - \exp \left[-\frac{eU_e}{ck_B T_e} A_y(x) + \frac{e\phi(x)}{k_B T_e} \right] \right\} \\ \frac{\partial^2 A_y}{\partial x^2} &= -\frac{4\pi e n_0}{c} \left\{ U_i \exp \left[\frac{eU_i}{ck_B T_i} A_y(x) - \frac{e\phi(x)}{k_B T_i} \right] - U_e \exp \left[-\frac{eU_e}{ck_B T_e} A_y(x) + \frac{e\phi(x)}{k_B T_e} \right] \right\} \cdot \hat{y} \end{aligned}$$

We assume no electric field and a quasineutral plasma, so with the same temperature for ions and electrons. We also assume the velocities for ions and electrons exactly opposite one to the other.

$$\phi = 0; \quad T_i = T_e = T; \quad U_i = -U_e = U$$

These conditions satisfy

$$\exp \left[\frac{eU_i}{ck_B T_i} A_y(x) \right] = \exp \left[-\frac{eU_e}{ck_B T_e} A_y(x) \right] \quad \Longleftrightarrow \quad \frac{U_i}{T_i} = -\frac{U_e}{T_e}$$

So the equation for the vector potential can be written as

$$\frac{d^2}{dx^2} A_y(x) = -\frac{8\pi e n_0}{c} U \exp \left[\frac{eU}{ck_B T} A_y(x) \right] \equiv -a e^{b A_y(x)}$$

with

$$a \equiv \frac{8\pi e n_0}{c} U; \quad b \equiv \frac{eU}{ck_B T}$$

Consider that A is a potential and it's defined up to a constant.

We choose as boundary condition

$$A(x=0) = 0; \quad B_z(x=0) = \left. \frac{\partial A_y}{\partial x} \right|_{x=0} = 0$$

We solve this ODE

$$\begin{aligned} A_y''(x) &= -a e^{b A_y} \\ \implies A_y'' A_y' &= -a A_y' e^{b A_y} \\ \implies \frac{1}{2} \left[(A_y')^2 \right]' &= -\frac{a}{b} \left(e^{b A_y} \right)' \\ \implies A_y'^2 &= -\frac{2a}{b} e^{b A_y} + C \end{aligned}$$

We impose the boundary condition for A_y'

$$0 = -\frac{2a}{b} + C \quad \implies \quad C = \frac{2a}{b}$$

So we return to the general solution

$$\begin{aligned}
 \Rightarrow A_y'^2 &= \frac{2a}{b} \left[1 - e^{bA_y} \right] \\
 \Rightarrow \frac{dA_y}{dx} &= \sqrt{\frac{2a}{b}} \left(1 - e^{bA_y} \right)^{1/2} \\
 \Rightarrow \frac{dA_y}{(1 - e^{bA_y})^{1/2}} &= \sqrt{\frac{2a}{b}} dx \\
 \Rightarrow \sqrt{\frac{2a}{b}} x &= \int_0^{A_y(x)} \frac{dA_y}{(1 - e^{bA_y})^{1/2}}
 \end{aligned}$$

Now we apply the substitution

$$t = e^{-bA_y/2}, \quad dt = -\frac{b}{2} e^{-bA_y/2} dA_y$$

so

$$dA_y = -\frac{2}{b} e^{bA_y/2} dt = -\frac{2}{b} \frac{1}{t} dt$$

and

$$\begin{aligned}
 \sqrt{\frac{2a}{b}} x &= \int_0^{\exp(-bA_y/2)} -\frac{2}{b} \frac{1}{t} \frac{dt}{(1 - \frac{1}{t^2})^{1/2}} = \\
 &= -\frac{2}{b} \int \frac{dt}{(t^2 - 1)^{1/2}} = \\
 &= -\frac{2}{b} \operatorname{arcosh} \left(e^{bA_y/2} \right)
 \end{aligned}$$

so

$$\begin{aligned}
 \sqrt{\frac{2a}{b}} x &= -\frac{2}{b} \operatorname{arcosh} \left[e^{bA_y(x)/2} \right] \\
 \Rightarrow \sqrt{\frac{ab}{2}} x &= -\operatorname{arcosh} \left[e^{bA_y(x)/2} \right] \\
 \Rightarrow \cosh \left(-\sqrt{\frac{ab}{2}} x \right) &= e^{-bA_y/2} \\
 \Rightarrow \ln \left[\cosh \left(\sqrt{\frac{ab}{2}} x \right) \right] &= -\frac{b}{2} A_y \\
 \Rightarrow A_y(x) &= -\frac{2}{b} \ln \left[\cosh \left(\sqrt{\frac{ab}{2}} x \right) \right]
 \end{aligned}$$

Now we substitute the constants a and b

$$\begin{cases} a \equiv \frac{8\pi en_0}{c} U \\ b \equiv \frac{eU}{ck_B T} \end{cases} \implies A_y(x) = -\frac{2ck_B T}{eU} \ln \left[\cosh \left(\sqrt{\frac{4\pi n_0 e^2 U^2}{c^2 k_B T}} x \right) \right]$$

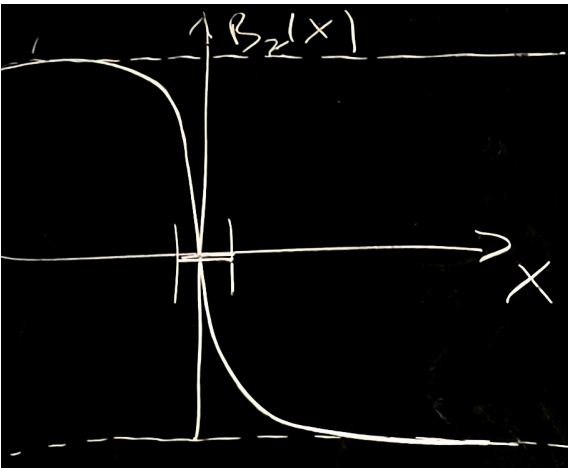
We isolate the Debye length λ_D

$$A_y(x) = -\frac{2ck_B T}{eU} \ln \left[\cosh \left(\frac{U}{c} \lambda_D^{-1} x \right) \right] \quad \text{with} \quad \lambda_D \equiv \sqrt{\frac{k_B T}{4\pi n_0 e^2}}$$

And the magnetic field will be

$$\begin{aligned} B_z(x) &= \frac{d}{dx} A_y(x) = \\ &= -\sqrt{16\pi n_0 k_B T} \cdot \tanh \left(\frac{U}{c\lambda_D} x \right) = \\ &\equiv -B_0 \cdot \tanh \left(\frac{U}{c\lambda_D} x \right) \end{aligned}$$

PIC B tanh



We find the density for ions and electrons using

$$n_\alpha(x) = \int f_\alpha d^3V = n_{\alpha 0} \exp \left[\frac{q_\alpha}{k_B T_\alpha} \left(\frac{1}{c} A_y(x) U_\alpha - \cancel{\phi(x)} \right) \right]$$

so

$$\begin{aligned} n_e = n_i &= n_0 \exp \left[\frac{q_\alpha U_\alpha}{ck_B T_\alpha} A_y(x) \right] = \\ &= n_0 \exp \left\{ \frac{\cancel{q_\alpha U_\alpha}}{\cancel{ck_B T_\alpha}} \left(-2 \frac{\cancel{ck_B T_\alpha}}{\cancel{eU}} \right) \ln \left[\cosh \left(\frac{U}{c\lambda_D} x \right) \right] \right\} = \\ &= n_0 \exp \left\{ -2 \ln \left[\cosh \left(\frac{U}{c\lambda_D} x \right) \right] \right\} = \\ &= n_0 \exp \left\{ \ln \left[\cosh^{-2} \left(\frac{U}{c\lambda_D} x \right) \right] \right\} = \\ &= \frac{n_0}{\cosh^2 \left(\frac{U}{c\lambda_D} x \right)} \end{aligned}$$

so

$$J = \sum_{\alpha} q_{\alpha} U_{\alpha} n_{\alpha} \simeq \cosh^{-2} \left(\frac{x}{\delta} \right)$$

We have a peak of density in the current sheath

PIC delta

